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Internal combustion engine for a motor vehicle, in particular for a motor car

Abstract

An internal combustion engine includes a suction tract, a compressor disposed in the suction tract, and an exhaust tract. A secondary air system is fluidically connected to the exhaust tract at an introduction point and to the suction tract at a first branching point downstream of the compressor where a part of air compressed by the compressor is branchable off from the suction tract and is introducible into the exhaust tract at the introduction point. The secondary air system is fluidically connected to the suction tract at a second branching point upstream of the compressor where a part of air is branchable off from the suction tract by the secondary air system and is introducible into the exhaust tract at the introduction point. Air branched off at the second branching point is feedable through the secondary air system and is feedable to the introduction point by a secondary air pump.

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Background/Summary

BACKGROUND AND SUMMARY OF THE INVENTION

(1) The invention relates to an internal combustion engine for a motor vehicle, in particular for a motor car.

(2) Such an internal combustion engine for a motor vehicle, in particular for a motor car, can for example be taken as known from DE 10 2007 057 603 A1. The internal combustion engine has a suction tract which can be flowed through by air, in which suction tract a compressor is arranged to compress the air flowing through the suction tract. In addition, the internal combustion engine has an exhaust tract, which can be flowed through by exhaust gas of the internal combustion engine, and a secondary air system, which is fluidically connected to the suction tract at at least one introduction point. The secondary air system is fluidically connected to the suction tract at a branching point arranged downstream of the compressor. At least a part of the air compressed by means of the compressor can be branched off from the suction tract at the branching point by means of the secondary air system and introduced into the secondary air system. The air branched off from the suction tract at the branching point and then introduced into the secondary air system can flow through the system, and is fed to the introduction point by means of the secondary air system, and introduced into the exhaust tract at the introduction point.

(3) DE 10 2013 226 063 A1 further discloses an internal combustion engine, having a combustion motor, a fresh gas train having a compressor and an exhaust train having an exhaust gas post-treatment device and a temperature sensor.

(4) The object of the invention is to develop an internal combustion engine of the kind specified in the introduction such that a particularly low-emission operation can be implemented.

(5) In order to develop an internal combustion engine of the kind specified herein such that a particularly low-emission operation can be implemented, it is provided according to the invention that the secondary air system is also fluidically connected to the suction tract at a second branching point arranged upstream of the compressor. At least a part of the air flowing through the suction tract upstream of the compressor can be branched off from the suction tract at the second branching point by means of the secondary air system and introduced into the secondary air system. The air branched off from the exhaust tract at the second branching point and introduced into the secondary air system can flow through the secondary air system, and is guided to the introduction point by means of the secondary air system, and introduced into the exhaust tract as secondary air. The secondary air system can thus branch off air from the suction tract both at the first branching point, and thus downstream of the compressor, and at the second branching point, and thus upstream of the compressor, and feed the air to the introduction point as secondary air, and introduce it into the exhaust tract at the introduction point.

(6) The secondary air system has a secondary air pump, by means of which the air branched off at the second branching point can be fed through the secondary air system and fed to the introduction point, and in particular fed into the exhaust tract. A particularly advantageous secondary air injection, which is particularly as required, can thus be implemented via the invention. The secondary air injection should be understood to mean that the air branched off at the respective

branching point can be introduced into the exhaust tract as secondary air at the introduction point, in particular while bypassing combustion chambers or all combustion chambers of the internal combustion engine. The air which is branched off from the suction tract at the first branching point, and thus downstream of the compressor, is also described as compressor air. The invention in particular makes it possible, for example, to introduce, i.e., to inject both the compressor air and the air branched off from the suction tract at the second branching point into the exhaust tract, for example from a particular point in time onwards. For example, the introduction point is arranged in an exhaust conduit, in particular in an outlet conduit, whereby the secondary air can be particularly advantageously injected into the exhaust tract. For example, the introduction point is arranged in an outlet conduit in particular formed or delimited by a cylinder head of the internal combustion engine. In particular, several, in particular all exhaust conduits, in particular outlet conduits, of the internal combustion engine can have a respective introduction point, at which the secondary air can be injected into the exhaust tract. The invention in particular makes it possible to introduce an advantageously large quantity of secondary air into the exhaust tract in all operating regions of the internal combustion engine, also described as a combustion motor, and preferably designed as a reciprocating piston motor, and thus to make a particularly low-emission operation of the internal combustion engine possible.

(7) Further advantages, features and details of the invention result from the following description of a preferred exemplary embodiment and with reference to the drawing. The features and combinations of features previously specified in the description and the features and combinations of features specified in the following description of the FIGURE and/or shown only in the single FIGURE can be used not only in the specified combination, but also in other combinations or in isolation, without leaving the scope of the invention.

Description

BRIEF DESCRIPTION OF THE DRAWING

(1) In the single FIGURE, the drawing shows a schematic depiction of an internal combustion engine for a motor vehicle, in particular for a motor car, having a secondary air system.

DETAILED DESCRIPTION OF THE DRAWING

(2) In a schematic depiction, the single FIGURE shows an internal combustion engine **10** for a motor vehicle, in particular for a motor car. The internal combustion engine **10** has an engine block **12**, also described as a cylinder block, by which cylinders **14** of the internal combustion engine **10** are formed. The respective cylinder **14** delimits a respective combustion chamber, in which combustion processes take place during a fired operation of the internal combustion engine **10**. Exhaust gas of the internal combustion engine **10** results from the respective combustion process. The internal combustion engine **10** has a suction tract **16** which can be flowed through by air, by means of which the air flowing through the suction tract **16** is fed to the combustion chambers and into the combustion chambers of the internal combustion engine **10**. The internal combustion engine has an exhaust gas turbocharger **18**, which has a compressor **20** arranged in the suction tract **16**, having a compressor wheel **22** arranged in the suction tract **16**. The air flowing through the suction tract **16** can be compressed by means of the compressor wheel **22**, and thus by means of the compressor **20**. Compressing the air flowing through the suction tract **16** is also described as charging, meaning that the compressed air is also described as charge air.

(3) The internal combustion engine **10** also has an exhaust tract **24**, which can be flowed through by the exhaust gas from the combustion chamber, and thus by the exhaust gas from the internal combustion engine **10**. The exhaust gas turbocharger **18** has a turbine **26** arranged in the exhaust tract **24**, the turbine having a turbine wheel **28** arranged in the exhaust tract **24**, which turbine wheel can be driven by means of the exhaust gas flowing through the exhaust tract **24**. The

compressor wheel **22** can be driven by the turbine wheel **28** via a shaft **30** of the exhaust gas turbocharger **18**, whereby the air flowing through the suction tract **16** can be compressed. An air filter **32** is arranged in the suction tract **16** upstream of the compressor **20**, i.e., upstream of the compressor wheel **22**, by means of which air filter the air flowing through the suction tract **16** is filtered. A throttle flap **34** is additionally arranged in the suction tract **16**, downstream of the compressor **20** and upstream of the combustion chambers, by means of which, for example, a respective quantity of the air flowing into the respective combustion chamber can be adjusted.

(4) An air recirculation system **36** having an air recirculation conduit **38** is assigned to the compressor **20**, which air recirculation conduit is fluidically connected to the suction tract **16** at connecting points **V1** and **V2**. By means of the air recirculation conduit **38**, at least a part of the air flowing through the suction tract **16** can be branched off from the suction tract **16** at a connecting point **V1** and introduced into the air recirculation conduit **38**. The air introduced into the air recirculation conduit **38** can flow through the air recirculation conduit **38**, and is fed to the introduction point **V2** by means of the air recirculation conduit **38**, and can flow into the suction tract **16** again at the connecting point **V2**. It can be seen that the connecting point **V2** is arranged downstream of the air filter **32** and upstream of the compressor wheel **22**, wherein the connecting point **V1** is arranged downstream of the compressor wheel **22** and upstream of the throttle flap **34**. The air recirculation system **36** additionally comprises an air recirculation valve **40**, which is arranged in the air recirculation conduit **38**. For example, a quantity of the air flowing through the air recirculation conduit **38** can be adjusted by means of the air recirculation valve **40**. In particular, the air recirculation valve **40** is designed as an electric air recirculation valve.

(5) A bypass device **42** is assigned to the turbine **26**, which bypass device has a bypass conduit **44**. The bypass conduit **44** is fluidically connected to the exhaust tract **24** at a connecting point **V3** and at a connecting point **V4**. By means of the bypass conduit **44**, at least a part of the exhaust gas flowing through the exhaust tract **24** can be branched off from the exhaust tract **24** at the connecting point **V3** and introduced into the bypass conduit **44**. The exhaust gas introduced into the bypass conduit **44** can flow through the bypass conduit **44**, and is fed to the connecting point **V4** by means of the bypass conduit **44**, and can flow back into the exhaust tract **24** at the connecting point **V4**. The exhaust gas flowing through the bypass conduit **44** bypasses the turbine **26**, and thus drives the turbine wheel **28**. The bypass conduit **44** is also described as a bypass or wastegate. The bypass device **42** comprises a bypass valve **46**, which is also described as a wastegate valve, and is arranged in the bypass conduit **44**. A quantity of the exhaust gas flowing through the bypass conduit **44** can be adjusted by means of the bypass valve **46**. It can be seen that the connecting point **V3** is arranged upstream of the turbine **26**, i.e., upstream of the turbine wheel **28**, the connecting point **V4** being arranged downstream of the turbine wheel **28**, i.e., downstream of the turbine **26**.

(6) An exhaust gas post-treatment device **48** is additionally assigned in the exhaust tract **24**, which exhaust gas post-treatment device is arranged downstream of the turbines **26**, in particular downstream of the connecting point **V4**. The exhaust gas post-treatment device **48** can be flowed through by the exhaust gas of the internal combustion engine, and can post-treat the exhaust gas. For this purpose, the exhaust gas post-treatment device for example comprises exhaust gas post-treatment elements **50a-c**. The exhaust gas post-treatment element **50a** is for example designed as a catalyst, in particular as a 3-way catalyst. The exhaust gas post-treatment element **50b** is for example designed as a particle filter, in particular as a petrol particulate filter (OPF). The exhaust gas post-treatment element **50c** is for example designed as a catalyst. In particular, the internal combustion engine also described as a combustion motor or motor can be designed as a petrol motor.

(7) The internal combustion engine **10** also has a secondary air system **52**. The secondary air system **52** is fluidically connected to the suction tract **16** at a first branching point **A1**. The secondary air system **52** is additionally fluidically connected to the exhaust tract **24** at introduction

points E. In particular, at least or exactly one introduction point E can be provided per combustion chamber, at which introduction point the secondary air system **52** is fluidically connected to the exhaust tract **24**. It is in particular conceivable that the introduction point E is arranged, in particular directly, in an exhaust conduit, in particular in an outlet conduit. For example, the outlet conduit is formed, i.e., delimited, by a cylinder head of the internal combustion engine **10**. The cylinder head is designed separately from the engine block **12**, and connected to the engine block **12**. In particular, the cylinder head forms a combustion chamber roof for the respective combustion chamber.

(8) It can be seen that the first branching point **A1** is arranged downstream of the compressor **20**, and thus downstream of the compressor wheel **22**. In the exemplary embodiment shown in the FIGURE, the first branching point **A1** is arranged downstream of the connecting point **V1** and upstream of the throttle flap **34**. The secondary air system **52**, at least a part of the air compressed by the compressor **20** can be branched off from the suction tract at the branching point **A1**, and can be introduced into the secondary air system **52**. Because the branching point **A1** is arranged downstream of the compressor wheel **22**, the air which is branched off from the suction tract **16** at the branching point **A1** by means of the secondary air system **52** is also described as compressor air. The compressor air can flow into the secondary air system **52**, and is fed to the respective introduction point E by means of the secondary air system **52**, and can flow out of the secondary air system **52** and into the exhaust tract **24** at the respective introduction point E. The compressor air is thus introduced, i.e., injected, into the exhaust tract **24** as secondary air at the respective introduction point E. Introducing the air branched off from the suction tract **16** at the branching point **A** and also described as compressor air or secondary air into the exhaust tract **24** at the respective introduction point E is also described as secondary air injection.

(9) To be able to implement a particularly low-emission operation of the internal combustion engine **10**, the secondary air system **52** is also fluidically connected to the suction tract **16** at a second branching point **A2**. It can be seen that the second branching point **A2** is arranged upstream of the compressor **20**, and thus upstream of the compressor wheel **22**, in particular upstream of the connecting point **V2**. By means of the secondary air system **52**, at least a part of the air flowing through the suction tract **16** upstream of the compressor **20** can be branched off from the suction tract **16** at the second branching point **A2**, and can be introduced into the secondary air system **52**. The air branched off from the suction tract **16** at the second branching point **A2** and introduced into the secondary air system **52** is also described as pre-compressor air, and can be fed to the respective introduction point E by means of the secondary air system **52**, and introduced, in particular injected, into the exhaust tract **24**, in particular as further secondary air, at the respective introduction point E. It can be seen that both the compressor air and the pre-compressor air is used as secondary air, which is branched off from the suction tract **16** at the respective branching point **A1** or **A2**, and is introduced into the suction tract **24** at the respective introduction point E. The respective introduction point E is arranged upstream of the exhaust gas post-treatment device **48**. In the exemplary embodiment depicted in the FIGURE, the respective introduction point E is arranged upstream of the turbine **26**, and in particular upstream of the connecting point **V3**.

(10) The secondary air system **52** has a first branch **Z1**, which is fluidically connected to the suction tract **16** at the second branching point **A2** and fluidically connected to the exhaust tract **24** at the respective introduction point E. It can be seen that the secondary air system **52** has a secondary air pump **54** which can in particular be operated electrically, by means of which air pump the pre-compressor air branched off at the second branching point **A2** can be fed through the secondary air system **52**, in particular through the first branch **Z1**, and fed to the respective introduction point E. The secondary air pump **54** is arranged in the first branch **Z1**.

(11) The secondary air system **52** additionally comprises a first check valve **56**, which is arranged downstream of the secondary air pump **54** in the first branch **Z1**. The check valve **56** closes in the direction of the secondary air pump **54**, and thus avoids a flow of air through the branch **Z1** in the

direction of the secondary air pump **54**. The check valve **56** opens in the direction of the respective introduction point E, however, meaning that the check valve **56** permits the flow of secondary air, in particular pre-compressor air, from the secondary air pump **54** to the respective introduction point E. The secondary air system **52** has a second branch **Z2**, which is fluidically connected to the suction tract **16** at the first branching point **A1**. The second branch **Z2** is fluidically connected to the first branch **Z1** at an opening point **M**. It can be seen that the opening point **M** is arranged downstream of the secondary air pump **54**, in particular downstream of the check valve **56**. The second branch **Z2** can thus be flowed through by the compressor air. A second check valve **58** of the secondary air system **52** is arranged in the second branch **Z2**. The second check valve **58** closes in the direction of the opening point **M** and opens in the direction of the branching point **A**.

(12) The secondary air system **52** additionally comprises a shutoff valve **60** provided in addition to the check valves **56** and **58**, which shutoff valve is arranged in the second branch **Z2**, and upstream of the check valve **58** and downstream of the branching point **A1**. The secondary air system **52** has a pressure sensor **62**, by means of which a pressure present in the secondary air system **52**, in particular downstream of the opening point **M**, can be detected. The secondary air system **52** further comprises a valve element **64**, which is in particular arranged in the branch **Z1**. The valve element **64** is presently arranged downstream of the opening point **M**, and in particular downstream of the pressure sensor **62**. The valve element **64** is for example a secondary air valve, in particular an exhaust gas combination valve. For example, a quantity of the secondary air which is introduced into the exhaust tract **24** at the respective introduction point E can be adjusted by means of the valve element **64**. It can be seen that the compressor air can mix with the pre-compressor air at the opening point **M**, and can thus form a total secondary air, wherein a quantity of the total secondary air can in particular be adjusted by means of the valve element **64**, which quantity should be or is introduced into the exhaust tract **24** at the respective introduction point E. In one embodiment, it would be conceivable that the check valve **58** opens in the direction of the opening point **M** and closes in the direction of the branching point **A1**, and thus, for example, permits a flow of the compressor air as secondary air from the branching point **A1** to the opening point **M**, and suppresses a reverse flow, i.e., a flow of air from the opening point **M** in the direction of the branching point **A1**.

LIST OF REFERENCE CHARACTERS

(13) **10** internal combustion engine **12** engine block **14** cylinder **16** suction tract **18** exhaust gas turbocharger **20** compressor **22** compressor wheel **24** exhaust tract **26** turbine **28** turbine wheel **30** shaft **32** air filter **34** throttle flap **36** air recirculation system **38** air recirculation conduit **40** air recirculation valve **42** bypass device **44** bypass conduit **46** bypass valve **48** exhaust gas post-treatment device **50a-c** exhaust gas post-treatment element **52** secondary air system **54** secondary air pump **56** check valve **58** check valve **60** shutoff valve **62** pressure sensor **64** valve element **A1** first branching point **A2** second branching point **E** introduction point **M** opening point **V1** connecting point **V2** connecting point **V3** connecting point **V4** connecting point **Z1** first branch **Z2** second branch

Claims

1. An internal combustion engine (**10**) for a motor vehicle, comprising: a suction tract (**16**) which is flowable through by air; a compressor (**20**) disposed in the suction tract (**16**) for compressing air flowing through the suction tract (**16**); an exhaust tract (**24**) which is flowable through by exhaust gas of the internal combustion engine (**10**); a turbine (**26**) disposed in the exhaust tract (**24**), wherein the turbine (**26**) drives the compressor (**20**); a secondary air system (**52**) which is fluidically connected to the exhaust tract (**24**) at an introduction point (E) and to the suction tract (**16**) at a first branching point (A1) disposed downstream of the compressor (**20**), wherein at the first branching point (A1) at least a part of air compressed by the compressor (**20**) is branchable off

from the suction tract (16) and is introducible into the exhaust tract (24) at the introduction point (E) as secondary air; wherein the secondary air system (52) is also fluidically connected to the suction tract (16) at a second branching point (A2) disposed upstream of the compressor (20), wherein at the second branching point at least a part of air flowing through the suction tract (16) upstream of the compressor (20) is branchable off from the suction tract (16) by the secondary air system (52) and is introducible into the exhaust tract (24) at the introduction point (E) as secondary air; and a secondary air pump (54), wherein air branched off at the second branching point (A2) is feedable through the secondary air system (52) and is feedable to the introduction point (E) by the secondary air pump (54); and wherein the introduction point (E) is upstream of the turbine (26).

2. The internal combustion engine (10) according to claim 1, wherein the secondary air system (52) has a first branch (Z1) which is fluidically connected to the suction tract (16) at the second branching point (A2) and to the exhaust tract (24) at the introduction point (E) and wherein the secondary air pump (54) is disposed in the first branch (Z1).

3. The internal combustion engine (52) according to claim 2, further comprising a first check valve (56) disposed in the first branch (Z1) downstream of the secondary air pump (54), wherein the first check valve (56) closes in a direction of the secondary air pump (54) and opens in a direction of the introduction point (E).

4. The internal combustion engine (10) according to claim 2, wherein the secondary air system (52) has a second branch (Z2) which is fluidically connected to the suction tract (16) at the first branching point (A1) and to the first branch (Z1) at an opening point (M).

5. The internal combustion engine (10) according to claim 4, further comprising a second check valve (58) disposed in the second branch (Z2), wherein the second check valve (58) opens in a direction of the first branching point (A1) and closes in a direction of the opening point (M).

6. The internal combustion engine (10) according to claim 5, wherein a shutoff valve (60) is disposed in the second branch (Z2) upstream of the second check valve.

7. The internal combustion engine (10) according to claim 4, further comprising a first check valve (56) disposed in the first branch (Z1) downstream of the secondary air pump (54), wherein the first check valve (56) closes in a direction of the secondary air pump (54) and opens in a direction of the introduction point (E) and wherein the opening point (M) is disposed downstream of the first check valve (56).

8. The internal combustion engine (10) according to claim 1, wherein the introduction point (E) is disposed in an outlet conduit that is assigned to a combustion chamber of the internal combustion engine (10) and that is formed by a cylinder head of the internal combustion engine (10).

9. An internal combustion engine (10) for a motor vehicle, comprising: a suction tract (16) which is flowable through by air; a compressor (20) disposed in the suction tract (16) for compressing air flowing through the suction tract (16); an exhaust tract (24) which is flowable through by exhaust gas of the internal combustion engine (10); a secondary air system (52) which is fluidically connected to the exhaust tract (24) at an introduction point (E) and to the suction tract (16) at a first branching point (A1) disposed downstream of the compressor (20), wherein at the first branching point (A1) at least a part of air compressed by the compressor (20) is branchable off from the suction tract (16) and is introducible into the exhaust tract (24) at the introduction point (E) as secondary air; wherein the secondary air system (52) is also fluidically connected to the suction tract (16) at a second branching point (A2) disposed upstream of the compressor (20), wherein at the second branching point at least a part of air flowing through the suction tract (16) upstream of the compressor (20) is branchable off from the suction tract (16) by the secondary air system (52) and is introducible into the exhaust tract (24) at the introduction point (E) as secondary air; and a secondary air pump (54), wherein air branched off at the second branching point (A2) is feedable through the secondary air system (52) and is feedable to the introduction point (E) by the secondary air pump (54); and wherein the introduction point (E) is disposed in an outlet conduit that is

assigned to a combustion chamber of the internal combustion engine (**10**) and the outlet conduit is formed in a cylinder head of the internal combustion engine (**10**).
